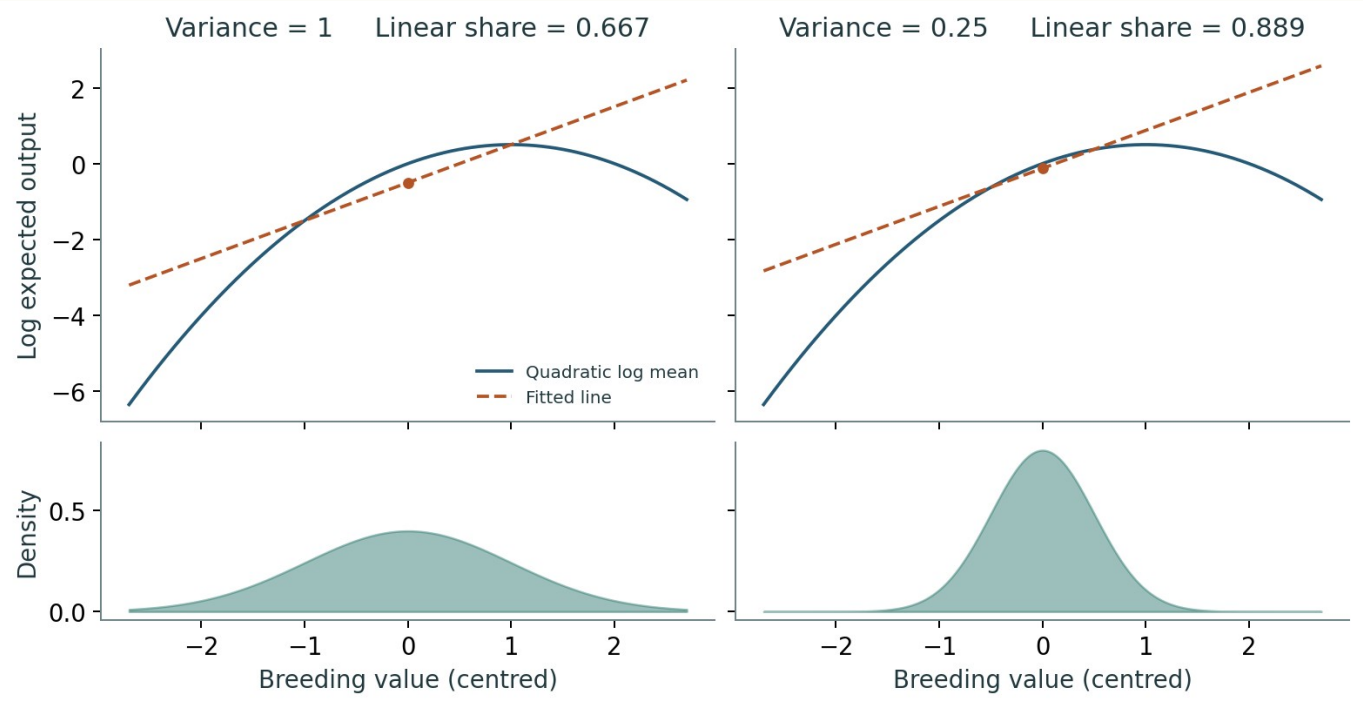
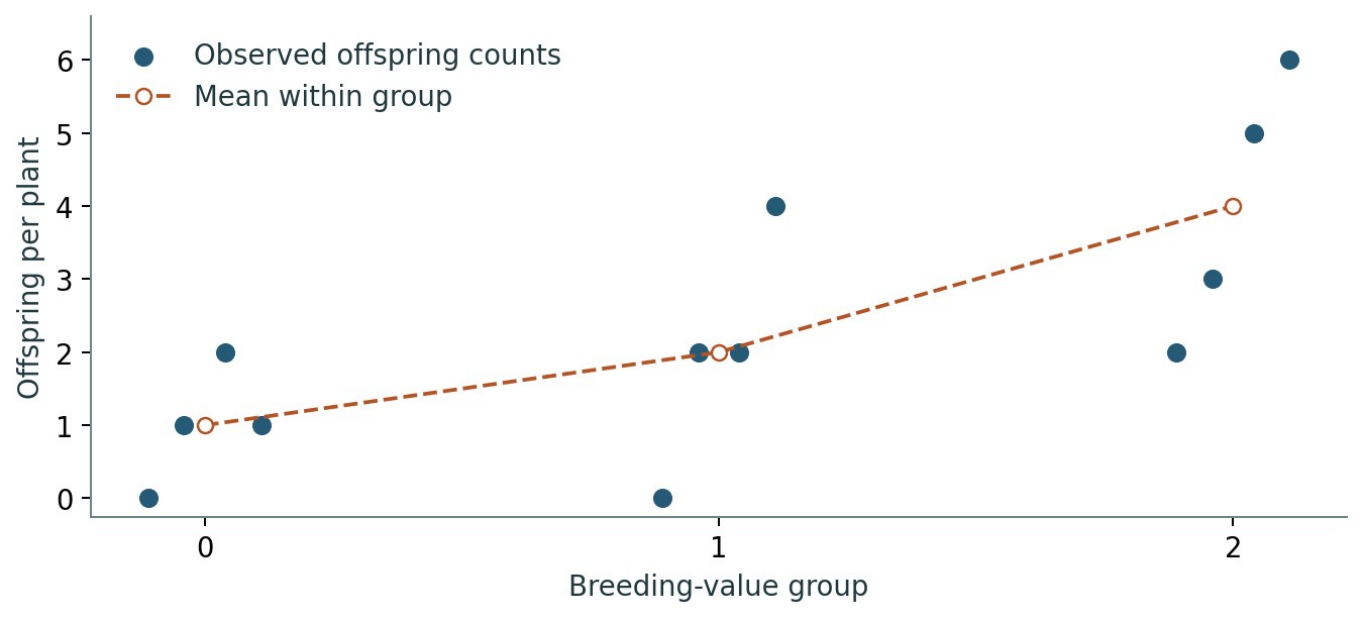
How much fitness variation can a straight line describe?



The same fitness curve can give different answers in two populations.

Compare the outcome at each breeding value with the frequency of that value.

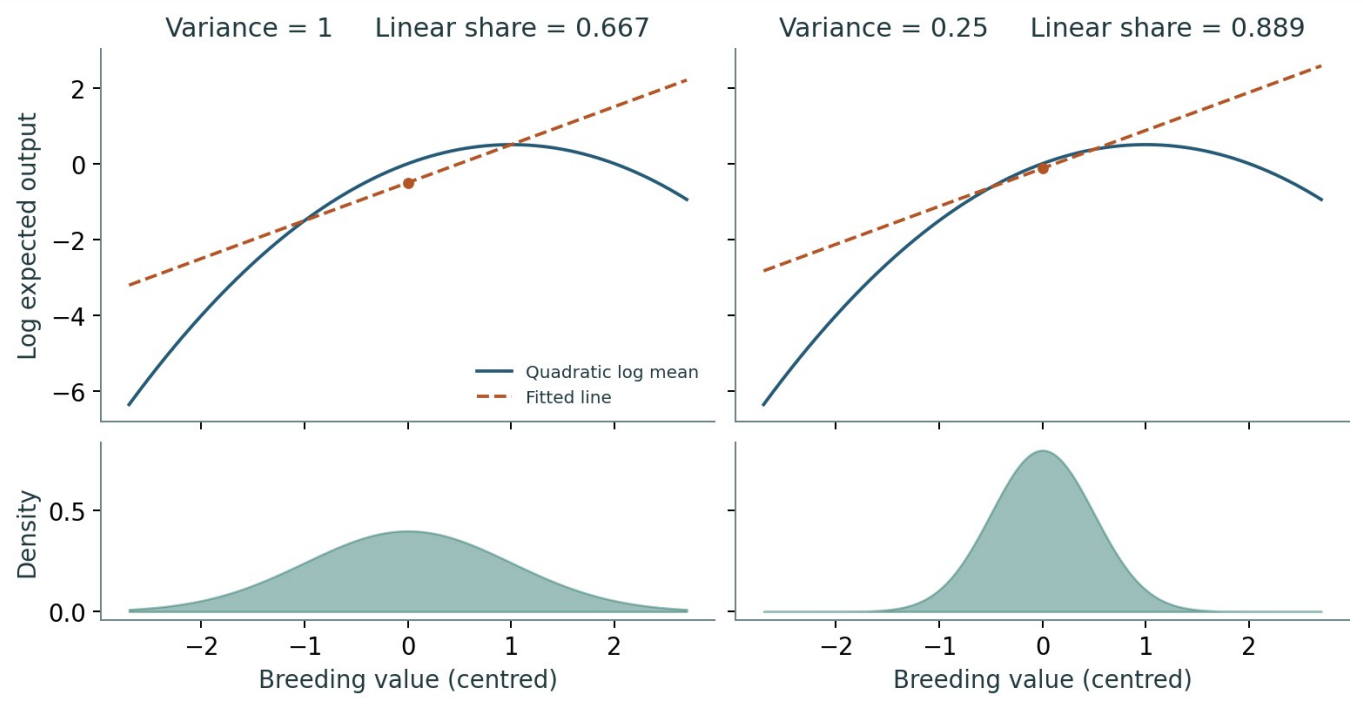
Average offspring counts before taking logs



Two plants with the same breeding value leave 0 and 4 surviving offspring. Their mean is 2.

A log of that mean exists. The average of their individual log counts does not.

Which breeding values occur most often?



Both populations have log expected output $y = x - x^2/2$. Their means agree. Only their variances differ.

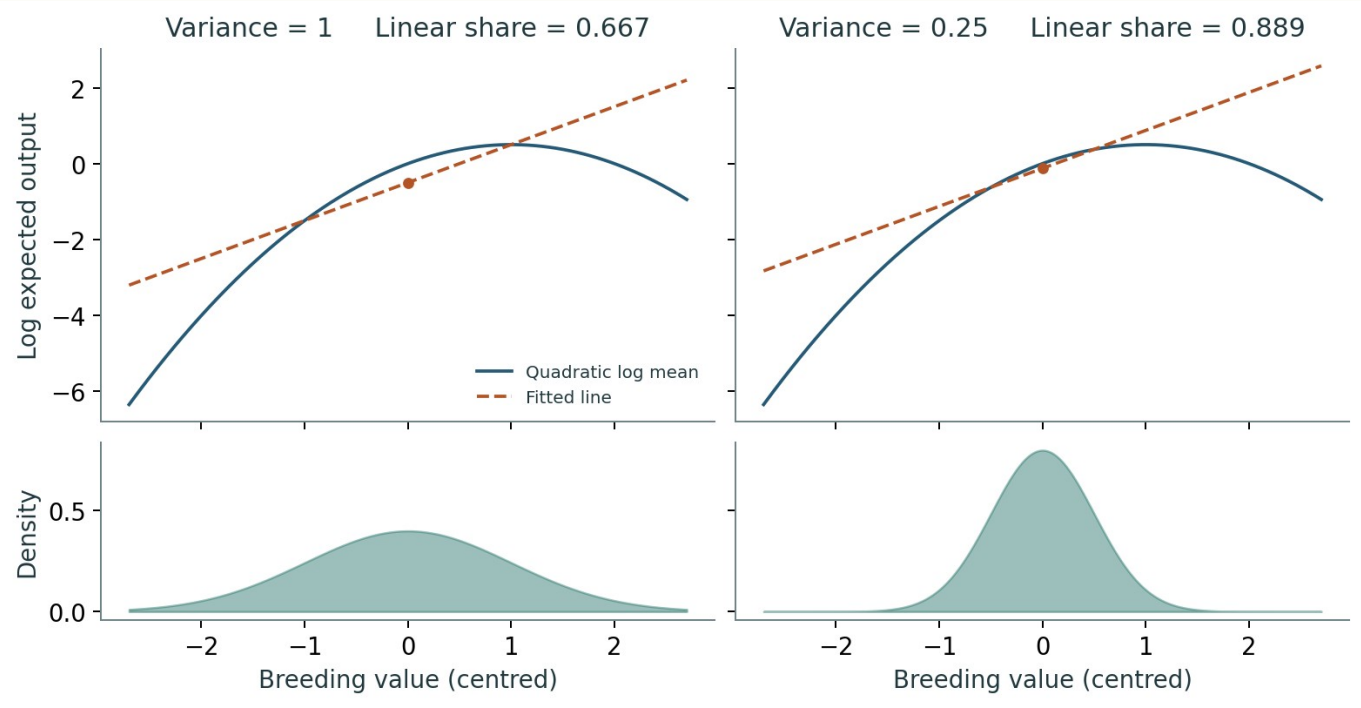
Read the density panel before comparing the fitted lines.

Slope contributes one factor of G; curvature contributes two

For a normal breeding value centred at zero, $L = bX$ and $Q = HX^2/2$. Symmetry gives $\mathrm{Cov}(L,Q) = 0$.

$V_{lin} = b^2G$
$V_{quad} = H^2G^2/2$

A larger share can accompany less linear variance



At G = 1, the line accounts for 2/3 of the variance. At G = 1/4, it accounts for 8/9.

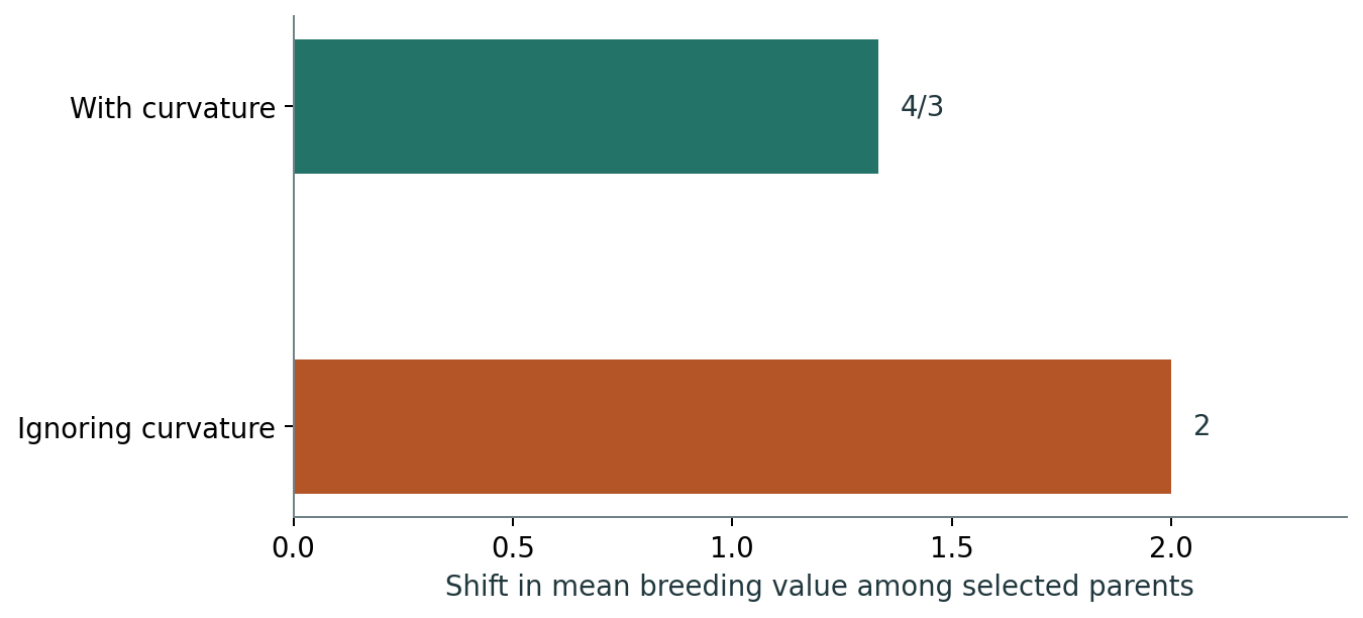
The linear variance fell fourfold. The quadratic variance fell sixteenfold.

Does a 97% fit bound the error in selection?

In the paper’s example, G = 1, slope b = 2 and curvature H = −1/2. The linear share is 4/(4 + 1/8).

Before showing the shift, ask which quantity selection uses as its weight.

The exact shift is 4/3; ignoring curvature gives 2



Multiply the original normal density by expected output. Completing the square gives a new mean of 4/3.

(2 − 4/3)/(4/3) = 50% overestimate